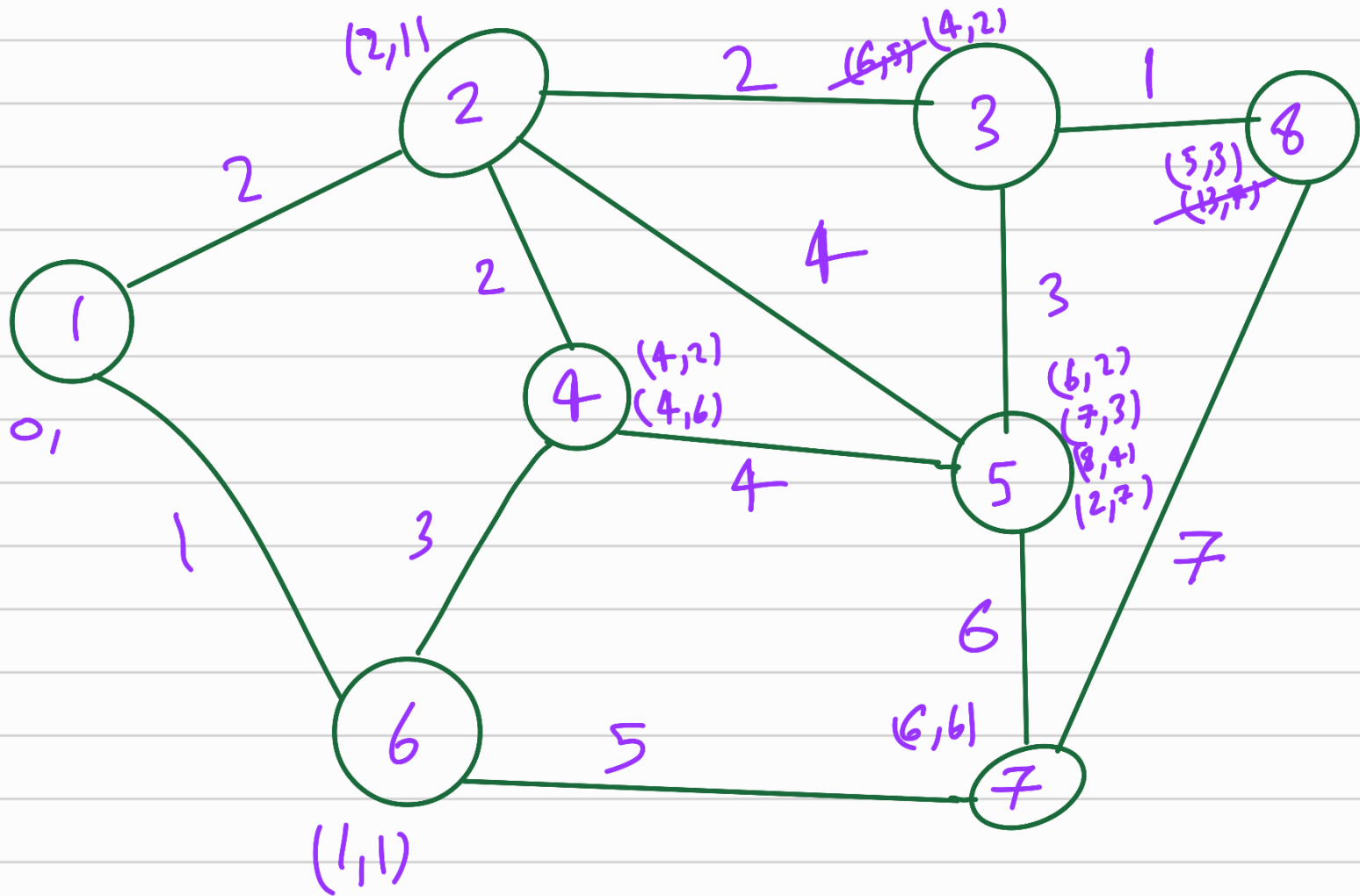


Name: Abdelrhman Mohamed Abdelhady Hodib

Id:2022513643

Operations research Assignment 2



Let $S = \{1\}$ $P_1 = 0$

connected to ① $P_6 = 1$ $P_2 = 2$

not connected to ① $P_3 = P_4 = P_5 = P_6 = P_7 = P_8 = \infty$

From 1 $S = \{1, 2\}$, $P_2 = 2$

From 2 $S = \{1, 2, 3, 4\}$, $P_3 = 4$

From 2 $S = \{1, 2, 3, 4\}$, $P_4 = 4$

6 From 2 $S = \{1, 2, 3, 4, 5\}$, $P_5 = \min\{8, 6, 7\}$

1 From 2 $S = \{1, 2, 3, 4, 5, 6\}$, $P_6 \min \{1, 7\}$

6 From 6 $S = \{1, 2, 3, 4, 5, 6, 7\}$, $P_7 \min \{6, 12\}$

5 From 3 $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $P_8 \min \{13, 5\}$

We will deduce that:

Shortest Length = 5

Shortest path = ① then ② then ③ then ⑧